Software Project Plan

Sample Application: Smart Library Management System

1. Project Overview

The Smart Library Management System is a web-based application that enables students and librarians to efficiently manage books, issue and return books, search the catalog, and track borrowing history. The system aims to reduce manual work and improve library operations.

2. Project Scope

In Scope

- User Registration and Login
- Book Management
- Book Search
- Book Issue and Return
- Fine Calculation
- Student Dashboard
- Admin Dashboard
- Reports Generation
- Notifications for Due Dates

Out of Scope

- Online Book Purchasing
- Mobile Application
- AI-Based Book Recommendation
- Integration with External Library Networks

3. Project Objectives

- Digitize library operations.
- Reduce manual record keeping.
- Improve book availability tracking.
- Enhance user experience.
- Generate accurate reports.

4. Project Timeline (10 Weeks)

Phase	Duration	Activities
Requirement Analysis	Week 1	Gather requirements and define scope
System Design	Week 2	UI Design, Database Design
Development	Weeks 3–6	Coding frontend and backend
Testing	Weeks 7–8	Unit Testing, Integration Testing
Deployment	Week 9	Deploy application and fix issues
Documentation & Presentation	Week 10	Prepare reports and final presentation

5. Team Roles

Role	Responsibilities
Project Manager	Planning, scheduling, monitoring progress
Business Analyst	Collect requirements and prepare documentation
UI/UX Designer	Design user interface and user experience

Role	Responsibilities
Frontend Developer	Develop web pages and user interface
Backend Developer	Develop server-side logic and APIs
Database Administrator	Design and maintain the database
QA/Test Engineer	Perform testing and report bugs
DevOps Engineer	Deployment and maintenance
Documentation Lead	Prepare technical and user documentation

6. Deliverables

- Software Requirement Specification (SRS)
- Project Plan
- System Design Document
- Source Code
- Database Schema
- Test Cases
- User Manual
- Final Project Report
- Presentation Slides

7. Resources Required

Hardware

- Laptop/Desktop
- Internet Connection

Software

- Visual Studio Code
- MySQL
- XAMPP
- Git & GitHub
- Figma
- Chrome Browser

8. Risk Management

Risk	Impact	Mitigation
Requirement changes	High	Conduct regular review meetings
Development delays	Medium	Follow project schedule and milestones
Team communication issues	Medium	Weekly progress meetings
Software bugs	High	Continuous testing and code review
Data loss	High	Regular backups using GitHub

9. Success Criteria

- All planned features are implemented.
- System operates without major defects.
- User acceptance testing is successful.
- Documentation is complete.
- Project is delivered on schedule.